

Computer Networks

IP Forwarding and Address Allocation

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IP forwarding

Package Forwarding in a Postal System

Postal analogy:

6 packages need to be sent in these addresses

- UP, Prayagraj, Barrister Mullah Colony, Swati
- UP, Kanpur, Kalyanpur, Amit
- West Bengal, Kharagpur, Inda, Manoj
- West Bengal, Durgapur, Mahatma Gandhi Rd, Prakash
- Maharashtra, Pune, Shivaji Nagar, Vivek
- Maharashtra, Nasik, Old Agra Road, Anand



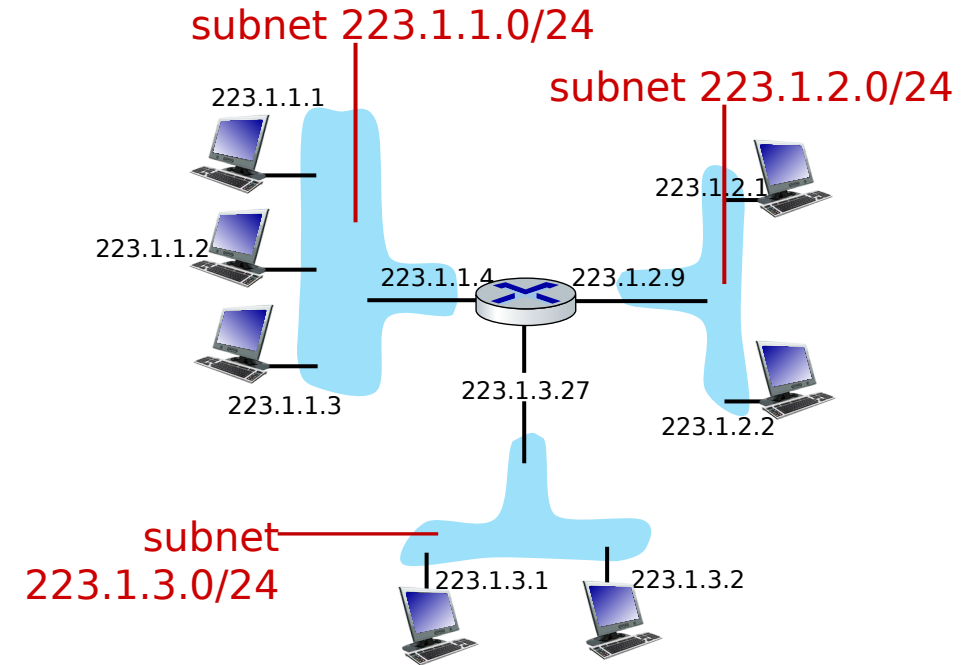
Src:
<https://pxhere.com/en/photo/1625663>

Destination prefix	Next hop
UP	Lucknow flight
West Bengal	Kolkata flight
Maharashtra	Mumbai flight

IP Forwarding

- IP addresses on a network belongs to the **same prefix**
- Nodes use a forwarding table that lists the prefixes and the corresponding next hop/link interface
 - Improves **scalability** → reduces the forwarding table size

Prefix	Next hop/interface
223.1.1.0/24	223.1.1.4
223.1.2.0/24	223.1.2.9
223.1.3.0/24	223.1.3.27



IP Forwarding

<i>forwarding table</i>	
Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010000 00000100 through 11001000 00010111 00010000 00000111	n 3
11001000 00010111 00011000 11111111	
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

But what happens if ranges don't divide up so nicely?

Longest Prefix Matching

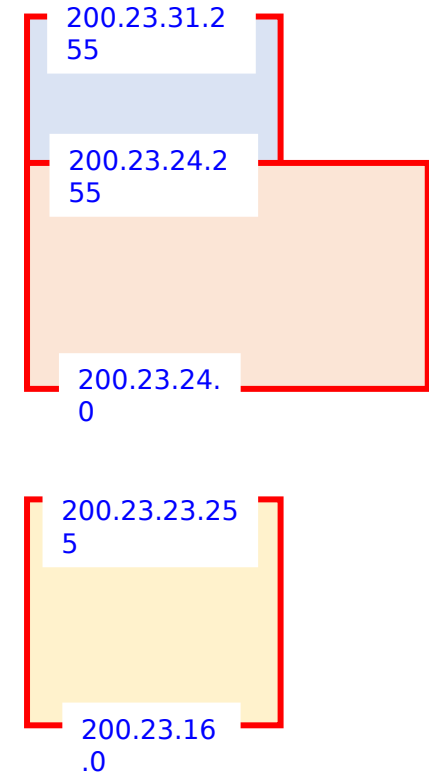
- Prefixes in a forwarding table might overlap

Longest prefix match

When looking for forwarding table entry for given destination address, use **longest** address prefix (**most specific entry**) that matches destination address

Destination Address Range	Link interface
*** ***** 11001000 00010111 00010	0
***** 11001000 00010111 00011000	1
*** ***** 11001000 00010111 00011	2
otherwise	3

Examples: 11001000 00010111 00010110 10100001 which interface?
11001000 00010111 00011000 10101010 which interface?



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11001000 00010111 00010110 10100001 *** *****	0
11001000 00010111 00011000 10101010 *****	1
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11001000 00010111 00011000 10101010 otherwise	3

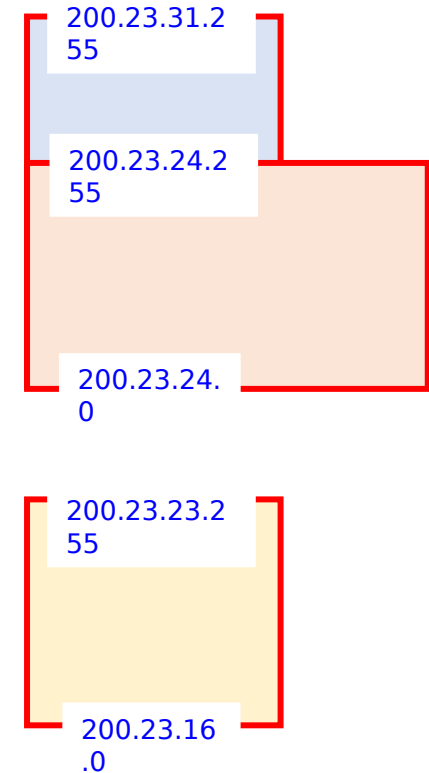
match!

Examples:

11001000 00010111 00010110 10100001
11001000 00010111 00011000 10101010

which interface?

which interface?



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*** *****	0
11001000 00010111 00010 *****	1
11001000 00010111 00011000 *** *****	2
11001000 00010111 00011 *****	3

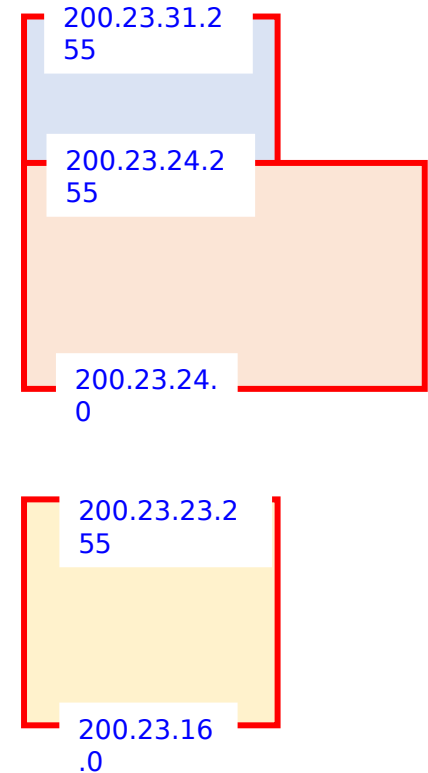
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Examples:

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11001000 00010111 00011000 10101010

which interface?

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Destination Address Range	Link interface
11001000 00010111 00010000 *** *****	0
11001000 00010111 00010000 *:*:*****	1
11001000 00010111 00010000 *** *****	2
11001000 00010111 00010000 *** *****	3

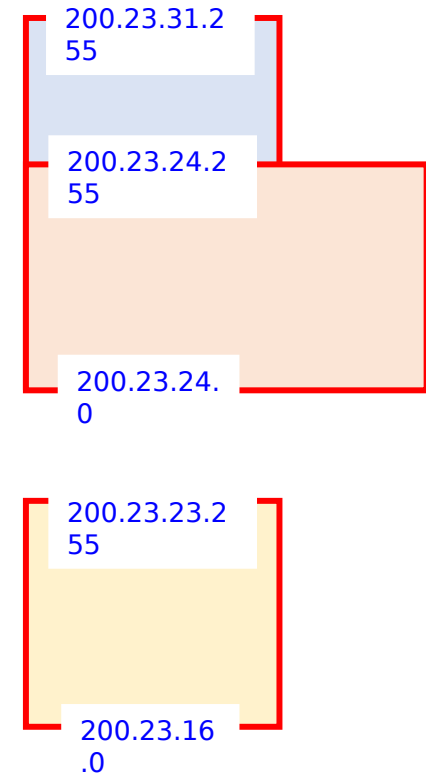
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Examples:

11001000 00010111 00010110 10100001
 11001000 00010111 00011000 10101010

which interface?

which interface?



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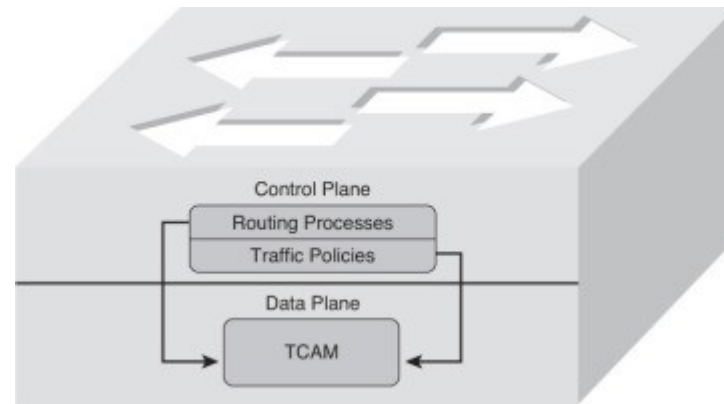


Src:
<https://pxhere.com/en/photo/1625663>

Destination prefix	Next hop
UP	Lucknow flight
UP, Prayagraj	Prayagraj flight
West Bengal	Kolkata flight
Maharashtra	Mumbai flight

Longest Prefix Matching

- Longest prefix matching: Often performed using **ternary content addressable memories (TCAMs)**
 - **Content addressable:** Present address to TCAM → retrieve address in one clock cycle, regardless of table size
 - Cisco Catalyst: ~1M routing table entries in TCAM



Src: <https://www.ciscopress.com/articles/article.asp?p=2264831&seqNum=2>

A diagram illustrating a central server or hub. A large, light blue, irregular shape represents the central system. Three desktop computers are connected to the left side of this shape by horizontal lines. A single horizontal line connects the right side of the central shape to a network router, which is depicted as a grey cylinder with a blue 'X' on its top surface.

- | | | |
|---|-----------------------|--------------------------------|
| 1 | | Broadcast on the local network |
| Network | 1 1 1 1 . . . 1 1 1 1 | Broadcast on a distant network |
| 127 | (Anything) | Loopback |

Prefix	Next hop
My network prefix	Send directly to that destination
0.0.0.0/0	Send to the router

```

Active Routes:
Network Destination        Netmask          Gateway           Interface         Metric
0.0.0.0                    0.0.0.0          172.27.16.254    172.27.17.157    291
127.0.0.0                  255.0.0.0        On-link           127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link           127.0.0.1        331
127.255.255.255           255.255.255.255  On-link           127.0.0.1        331
172.27.0.0                 255.255.0.0      On-link           172.27.17.157    291
172.27.17.157             255.255.255.255  On-link           172.27.17.157    291
172.27.255.255            255.255.255.255  On-link           172.27.17.157    291
224.0.0.0                  240.0.0.0        On-link           127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link           172.27.17.157    291
255.255.255.255           255.255.255.255  On-link           127.0.0.1        331
255.255.255.255           255.255.255.255  On-link           172.27.17.157    291

```

How To Obtain IP Address?

IP Addresses: How to Get One?

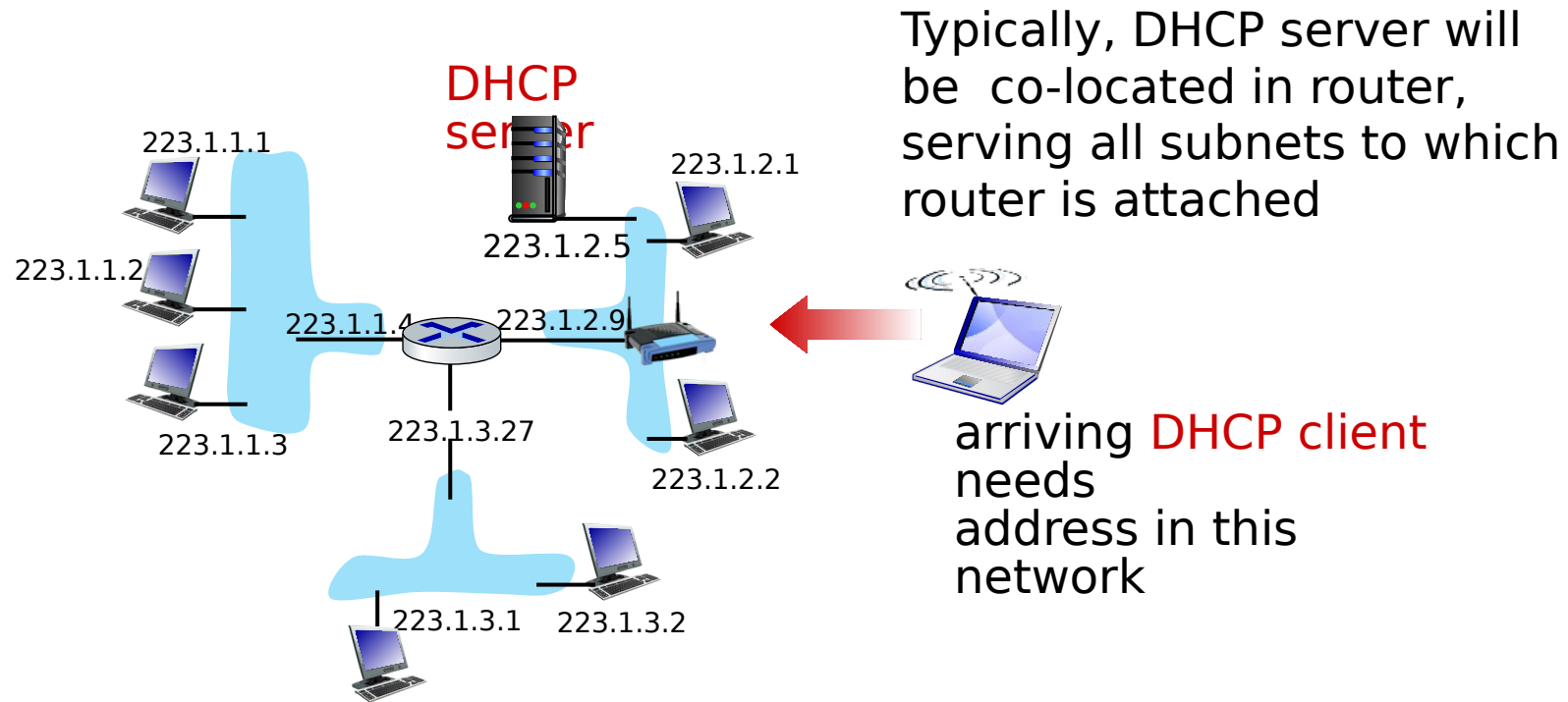
That's actually **two** questions:

1. How does a host get IP address within its network (host part of address)?
2. How does a network get IP address for itself (network part of address)

How does host get IP address?

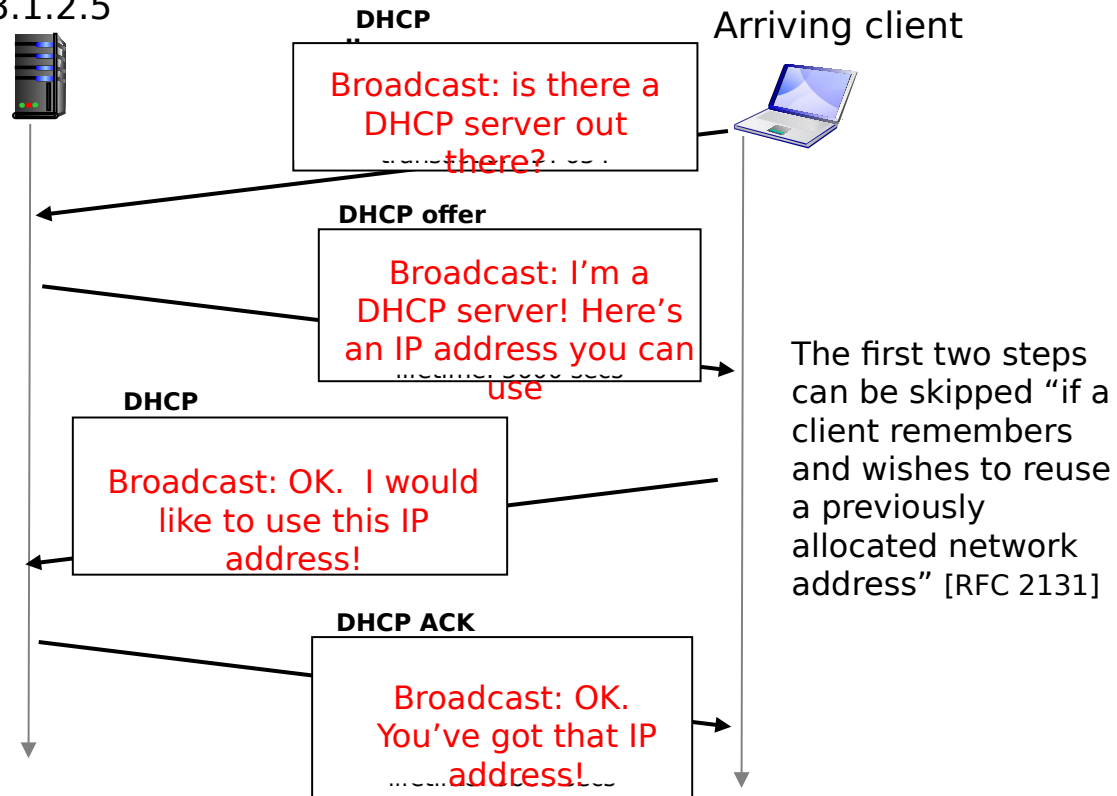
- Hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
 - **DHCP: Dynamic Host Configuration Protocol**: dynamically get address from a server
 - "Plug-and-play"
-

DHCP Client-Server Scenario



DHCP Client-Server Scenario

DHCP server: 223.1.2.5



DHCP: Dynamic Host Configuration Protocol

DHCP overview:

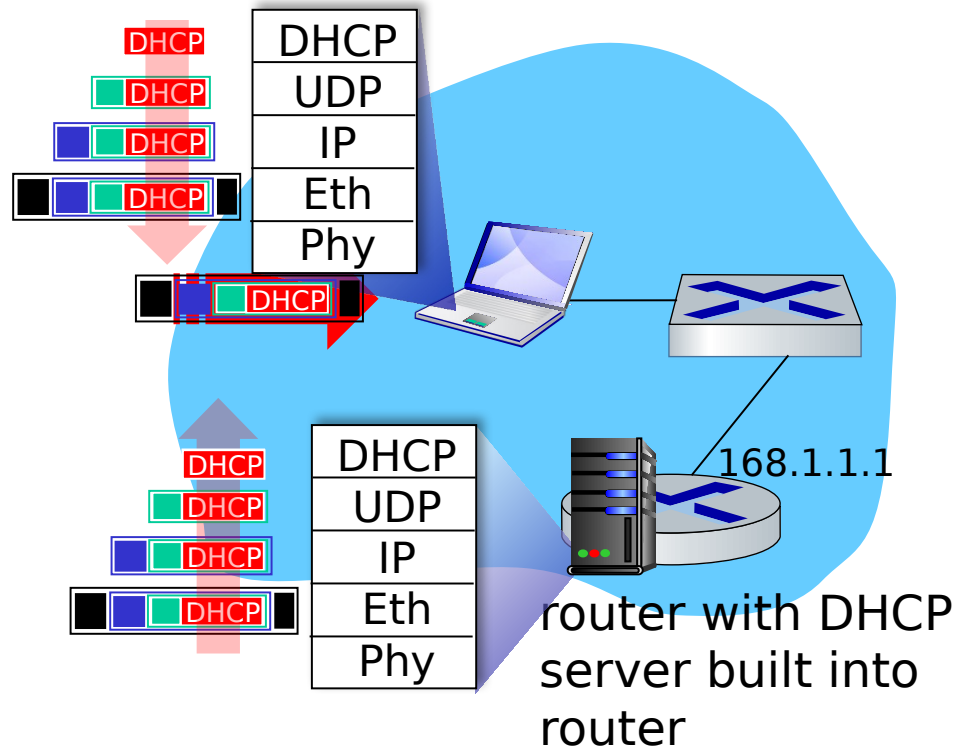
- Host broadcasts **DHCP discover** msg [optional]
 - DHCP server responds with **DHCP offer** msg [optional]
 - Host requests IP address: **DHCP request** msg
 - DHCP server sends address: **DHCP ack** msg
-

DHCP: More Than IP Addresses

DHCP can return more than just allocated IP address on subnet:

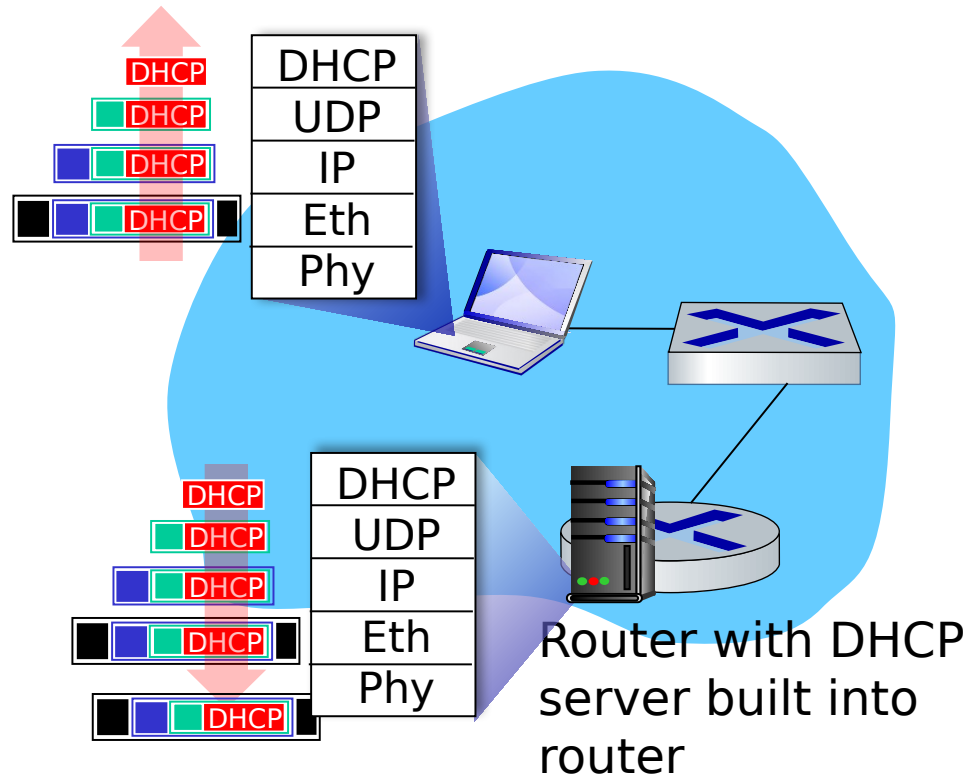
- Address of first-hop router for client
 - Name and IP address of DNS sever
 - Network mask (indicating network versus host portion of address)
-

DHCP Example



- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP

DHCP Example



- DHCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- Encapsulated DHCP server reply forwarded to client, demuxing up to DHCP at client
- Client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

How Does Networks Get IP addresses?

Q: How does network get subnet part of IP address?

A: Gets allocated portion of its provider ISP's address space

ISP's block 11001000 00010111 00010000 00000000
200.23.16.0/20

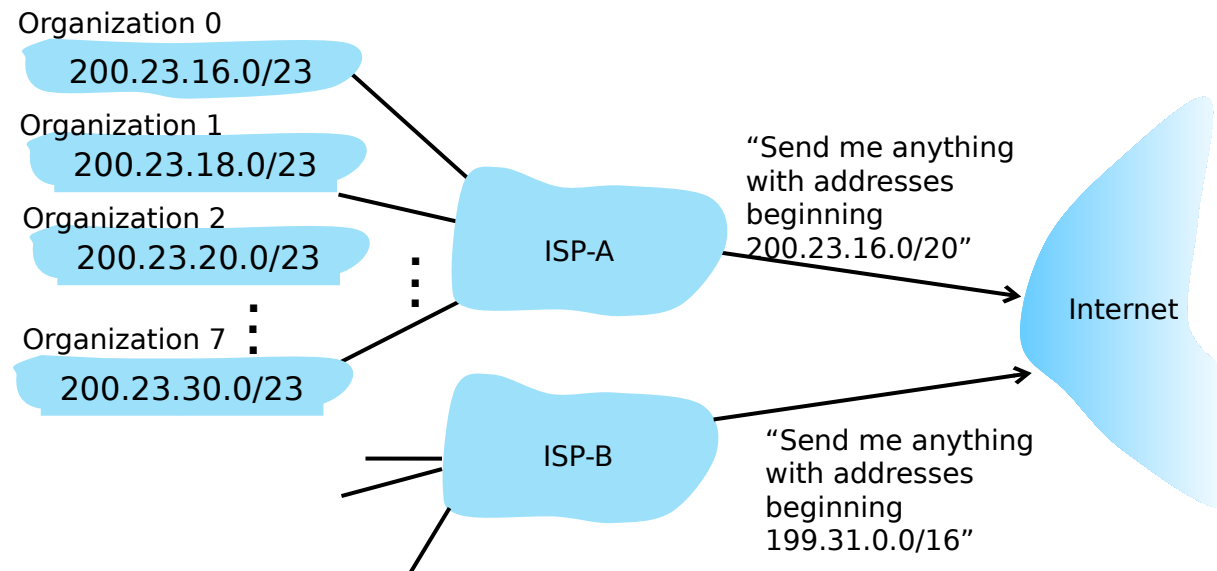
ISP can then allocate out its address space in 8 blocks:

Organization 0	<u>11001000</u>	<u>00010111</u>	<u>00010000</u>	00000000
200.23.16.0/23				
Organization 1	<u>11001000</u>	<u>00010111</u>	<u>00010010</u>	00000000
200.23.18.0/23				
Organization 2	<u>11001000</u>	<u>00010111</u>	<u>00010100</u>	00000000
200.23.20.0/23				

...		
Organization 7	<u>11001000</u>	<u>00010111</u> <u>00011110</u> <u>00000000</u>
200.23.30.0/23		

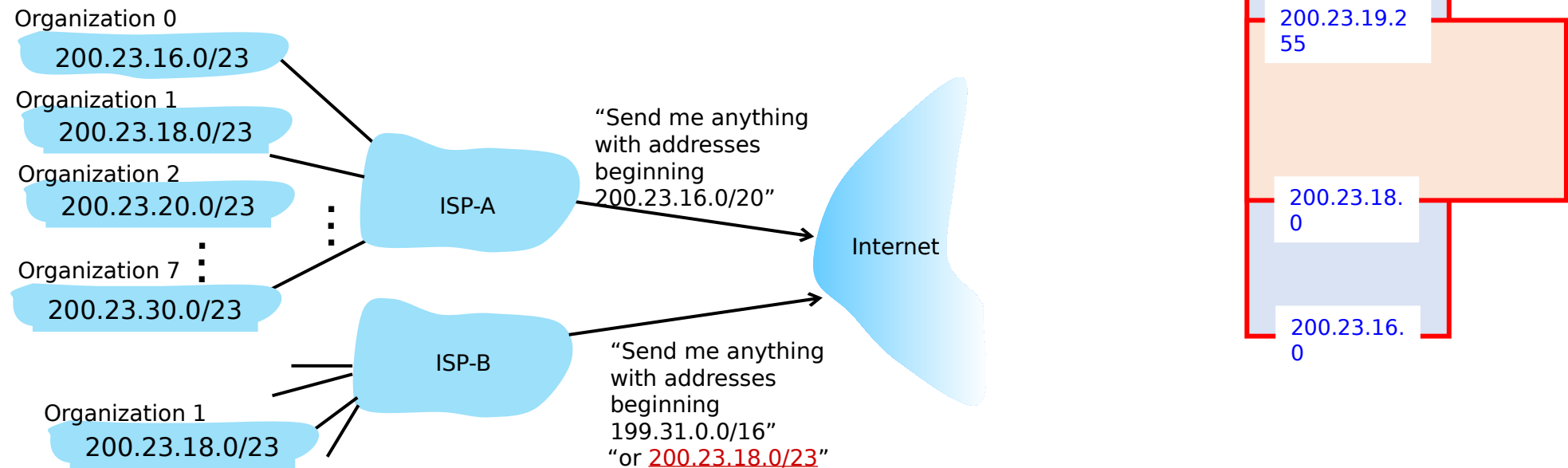
Hierarchical Addressing: Route Aggregation

Hierarchical addressing allows efficient advertisement of routing information:



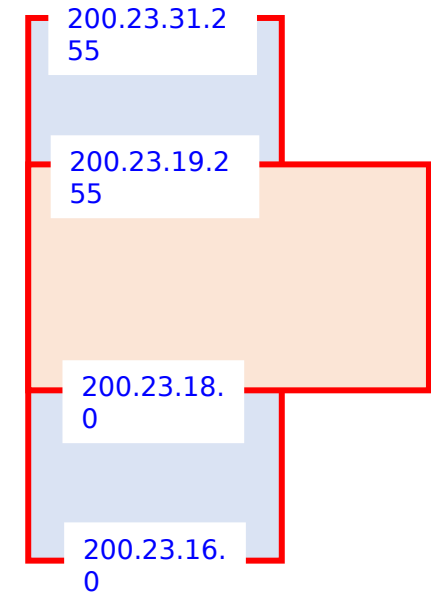
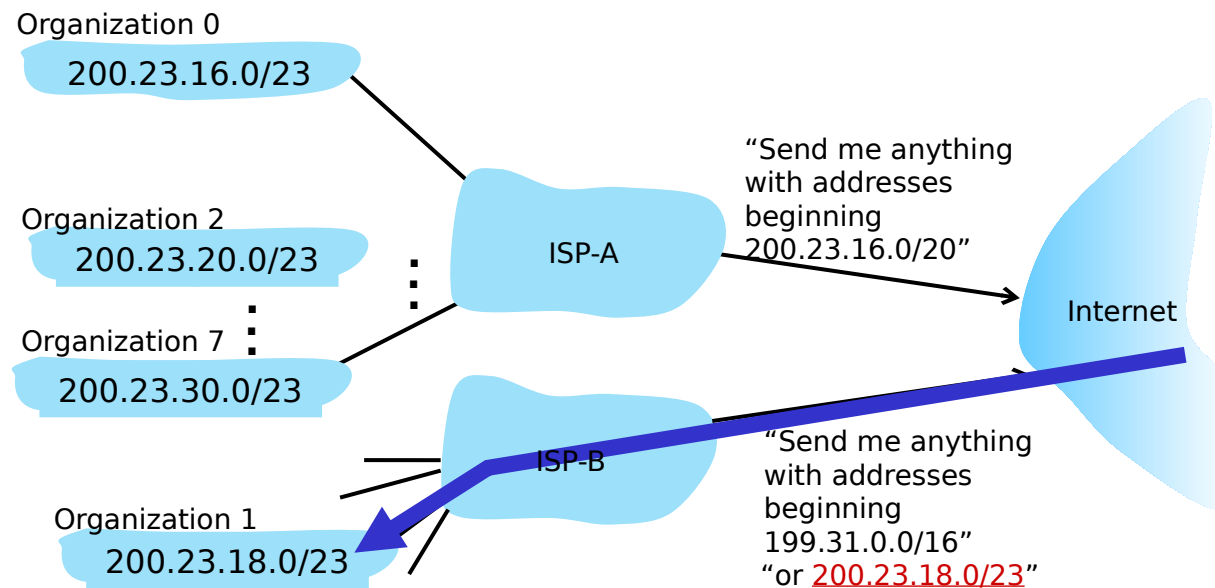
Hierarchical Addressing: More Specific Routes

- Organization 1 moves from ISP-A to ISP-B
- ISP-B now advertises a **more specific route** to Organization 1



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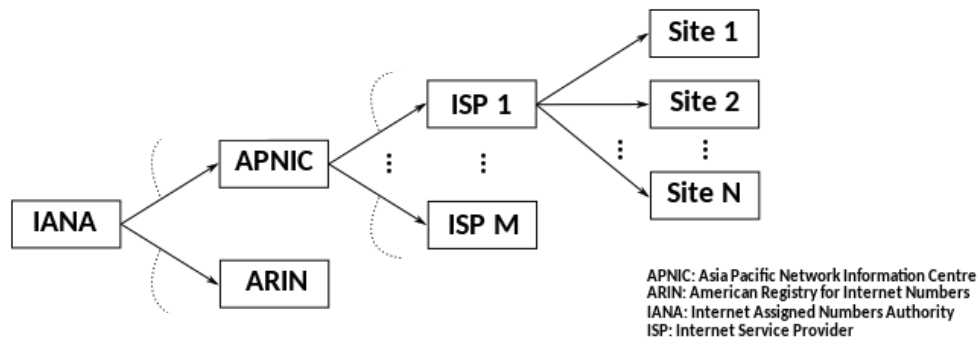


How ISPs Get IP Addresses?

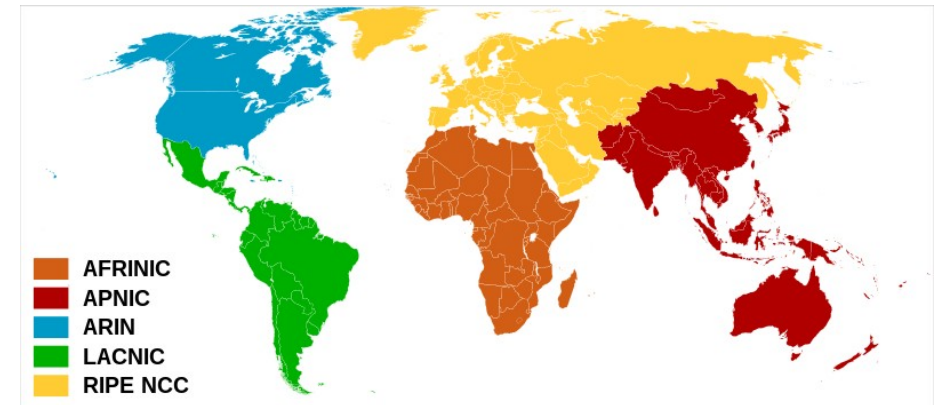
ICANN: Internet Corporation for Assigned Names and Numbers

<http://www.icann.org/>

- Allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- Manages DNS root zone, including delegation of individual TLD (.com, .edu , ...) management



Src: https://commons.wikimedia.org/wiki/File:IPv6_Prefix_Assignment_Example-en.svg

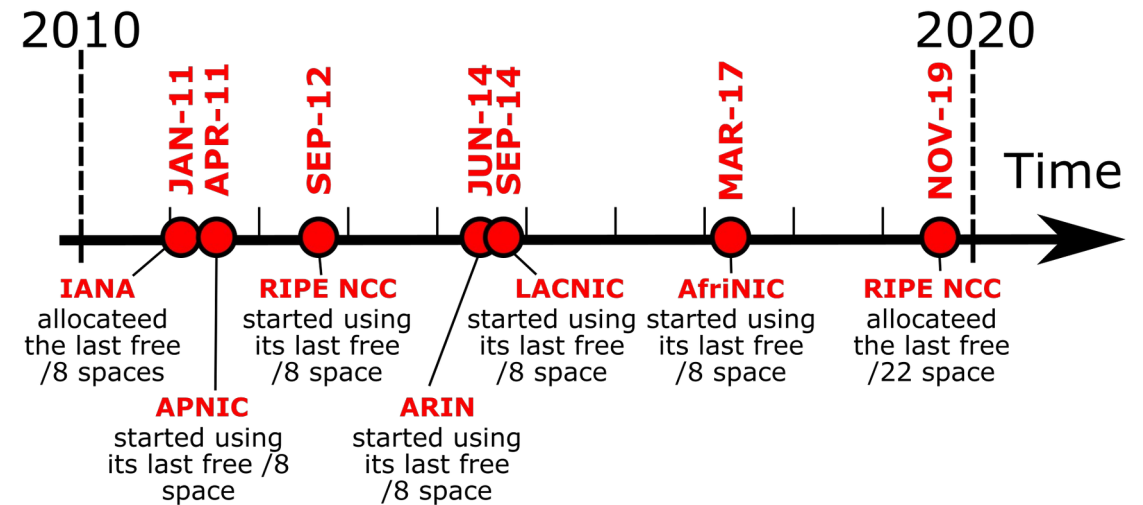


Src: https://commons.wikimedia.org/wiki/File:Regional_Internet_Registries_world_map.svg

IPv4 Addressing: Concluding Remarks

Are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT helps IPv4 address space exhaustion
- IPv6 has 128-bit address space



Src:
https://commons.wikimedia.org/wiki/File:IPv4_IANA_and_RIR_exhaustion_time_line-en.svg